16. The value of the series $\frac{1}{2}+\frac{7}{4}+\frac{13}{8}+\frac{19}{16}+\cdots$ is

(1) 14 (2) 7 (3) 4 (4) 6.

17. The sum of an infinite GP is 18. If the first term is 6, the common ratio is

(1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{1}{6}$ (4) $\frac{3}{4}$.

18. The coefficient of x^5 in the series e^{-2x} is

(1) $\frac{2}{3}$ (2) $\frac{3}{2}$ (3) $\frac{-4}{15}$ (4) $\frac{4}{15}$.

19. The value of $\frac{1}{2!}+\frac{1}{4!}+\frac{1}{6!}+\cdots$ is

(1) $\frac{e^2+1}{2e}$ (2) $\frac{(e+1)^2}{2e}$ (3) $\frac{(e-1)^2}{2e}$ (4) $\frac{e^2-1}{2e}$.

20. The value of $1-\frac{1}{2}\left(\frac{2}{3}\right)+\frac{1}{3}\left(\frac{2}{3}\right)^2-\frac{1}{4}\left(\frac{2}{3}\right)^3+\cdots$ is

(1) $\log\left(\frac{5}{3}\right)$ (2) $\frac{3}{2}\log\left(\frac{5}{3}\right)$ (3) $\frac{5}{3}\log\left(\frac{5}{3}\right)$ (4) $\frac{2}{3}\log\left(\frac{2}{3}\right)$.

Summary

In this chapter we have acquired the knowledge of

- Binomial theorem for any $n \in \mathbb{N}$,

$$(a+b)^n = {}^n\mathrm{C}_0\, a^n b^0 + {}^n\mathrm{C}_1\, a^{n-1}b^1 + \cdots + {}^n\mathrm{C}_n\, a^0 b^n.$$

- ${}^n\mathrm{C}_0 + {}^n\mathrm{C}_1 + \cdots + {}^n\mathrm{C}_n = 2^n$.
- ${}^n\mathrm{C}_1 + {}^n\mathrm{C}_3 + + {}^n\mathrm{C}_5 \cdots + = {}^n\mathrm{C}_0 + {}^n\mathrm{C}_2 + + {}^n\mathrm{C}_4 \cdots + = 2^{n-1}$
- $AM \geq GM \geq HM$
- The nth term of an AP is given by $T_n = a + (n-1)d$.
- The nth term of an GP is given by $T_n = ar^{n-1}$.
- The nth term of an AGP is given by $T_n = (a+(n-1)d)r^{n-1}$.
- For any positive numbers a and b, we have

$$\text{AM} = \frac{a+b}{2},\ \text{GM} = \sqrt{ab},\ \text{HM} = \frac{2ab}{a+b}.$$

- The sum of first n terms of an AP is given by $S_n = \frac{n}{2}(2a+(n-1)d)$.
- The sum of first n terms of an GP is given by $S_n = \frac{a(1-r^n)}{1-r}$ for $r \neq 1$.
- The sum of first n terms of an AGP is given by $S_n = \frac{a-(a+(n-1)d)r^n}{1-r} + dr\left(\frac{1-r^{n-1}}{(1-r)^2}\right)$ for $r \neq 1$.
- $\sum_{k=1}^{n} k = 1+2+3+\cdots+n = \frac{n(n+1)}{2}$.
- $\sum_{k=1}^{n} k^2 = 1^2+2^2+3^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}$.